

DATA1901 Realistic Practice Exam

Original exam-style paper with answer key and worked solutions

2026-06-10

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DATA1901 Realistic Practice Exam

This is an original practice exam generated from the DATA1901 project materials in this workspace. It mirrors the structure and assessment style signalled by the local exam guide and sample/practice exams: interpretation-heavy multiple choice questions, R output, data stories, and one extended response.

Primary local sources used:

- GPT_Project_Memory_MD/15_Exam_Preparation_and_Guides.md
- GPT_Project_Memory_MD/16_Sample_Exams_Practice_and_Answers.md
- GPT_Project_Memory_MD/01_Week_01_Topic_1_and_Mini_Lecture.md to 11_Week_11_Topic_11_and_Mini_Lecture.md
- GPT_Project_Memory_MD/13_Coding_Milestones_01-11.md
- GPT_Project_Memory_MD/14_ggplot2_Reference.md
- DATA1901_Mastery_Notes/Topic_01_Design_of_Experiments.md to Topic_11_Tests_for_Relationships.md

This is not an official University of Sydney examination paper.

Instructions

Time allowed: 10 minutes reading time, 2 hours writing time.

The exam has two sections:

Section	Questions	Suggested weighting
Multiple Choice	30 questions	75%
Extended Answer	1 data-story question with 5 parts	25%

Choose at most one answer for each multiple choice question. For the extended answer, show reasoning and write conclusions in context.

Do not read the worked solutions until you have attempted the paper.

Section A: Multiple Choice

In each question, choose the most appropriate option.

General DATA1901 Concepts

1. Data provenance means:

- ☒ (a) A record of where data came from, how it was collected, and how it has been processed
- ☐ (b) The probability that the null hypothesis is true
- ☐ (c) The process of choosing colours for a graph
- ☐ (d) The number of rows in a dataset

2. Which study is most clearly a randomised controlled trial?

- ☐ (a) Comparing people who already use a sleep app with people who do not
- ☒ (b) Randomly assigning consenting students to either a study-planning app or usual study habits
- ☐ (c) Surveying students about whether they like recorded lectures
- ☐ (d) Analysing five years of rainfall data from weather stations

3. A researcher takes a random sample of adults from Sydney and records whether each person owns an electric vehicle. No treatment is assigned. Which conclusion is best supported?

- ☐ (a) The study can prove electric vehicle ownership causes lower emissions
- ☒ (b) The study can estimate features of the Sydney adult population, but it cannot by itself prove causation
- ☐ (c) The study is a randomised controlled trial because the sample was random
- ☐ (d) The study has no value because no treatment was assigned

4. Which graphical summary is most suitable for one qualitative variable and one quantitative variable?

- ☐ (a) Histogram
- ☐ (b) Scatterplot
- ☒ (c) Comparative boxplot
- ☐ (d) Q-Q plot



5. A tidy dataset means:

- ☐ (a) The dataset has no missing values
- ☒ (b) Each variable is a column and each observation is a row
- ☐ (c) The data has been sorted alphabetically
- ☐ (d) Every variable is quantitative

6. A quantitative variable x has mean 12 and standard deviation 5. A new variable is defined by $y = 4 - 0.5x$. What are the mean and standard deviation of y ?

- ☐ (a) Mean = 10, SD = -2.5
- ☐ (b) Mean = -2, SD = -2.5
- ☐ (c) Mean = 10, SD = 2.5
- ☒ (d) Mean = -2, SD = 2.5

$$\begin{aligned} \mu &= 12 & \sigma &= 5 \\ y &= 4 - 0.5x & \bar{y} &= 4 - 0.5(12) \\ & & &= 4 - 6 \\ & & &= -2 \\ & & 10.5 \times 5 & 2.5 \times 2 \end{aligned}$$

7. A dataset has $Q1 = 18$ and $Q3 = 30$. Using the boxplot rule, which value would be considered a high outlier?

- ☐ (a) 45
- ☐ (b) 48
- ☒ (c) 49
- ☐ (d) 30

$$\begin{aligned} 12 &= IQR \\ 30 + 1.5 \times 12 &= 30 + 18 \\ &= 48 \end{aligned}$$

8. If $X \sim N(80, 4^2)$, which R command calculates $P(72 < X < 84)$?

$$\sigma = 4 \quad \mu = 80 \quad 3$$

$$p(84, 80, 4) - p(72, 80, 4)$$

- ☒ (a) `pnorm(84) - pnorm(72)`
- ☒ (b) `pnorm(84, 80, 4) - pnorm(72, 80, 4)`
- ☒ (c) `qnorm(84, 80, 4) - qnorm(72, 80, 4)`
- ☒ (d) `pnorm(72, 80, 4) + pnorm(84, 80, 4)`

9. Measurement error can be written as:

$$\text{measurement} = \text{exact value} + \text{chance error} + \text{bias}$$

How could the chance error component be estimated?

- ☒ (a) Remove all outliers and calculate the mean
- ☒ (b) Increase the sample size until the result is correct
- ☒ (c) Estimate the exact value using a boxplot
- ☒ (d) Repeat measurements under the same conditions and calculate the standard deviation

* 10. Two variables have correlation $r = -0.60$. New variables are created by $\text{new_x} = 10 - 2x$ and $\text{new_y} = 3 + 4y$. What is the correlation between new_x and new_y ?

- ☒ (a) -0.60
- ☒ (b) -0.30
- ☒ (c) 0.60
- ☒ (d) 0.30

$$\begin{aligned} x &= 10 - 2x \\ y &= 3 + 4y \end{aligned} \quad \begin{aligned} 3x &= 10 \quad x = \frac{10}{3} \\ y &= -1 \end{aligned}$$

11. Consider the R output:

```
lm(energy_use ~ temperature)
```

Coefficients:

(Intercept) temperature
31.50 -0.42

$$\text{energy_use} = a + b \text{temp.}$$

Which interpretation is most appropriate?

- ☒ (a) For each one-unit increase in temperature, predicted energy use decreases by 0.42 units
- ☒ (b) For each one-unit increase in temperature, predicted energy use decreases by 31.50 units
- ☒ (c) The correlation is -0.42
- ☒ (d) The mean energy use is -0.42

12. In linear regression, a residual is:

- ☒ (a) Predicted value minus observed value
- ☒ (b) Observed value minus predicted value
- ☒ (c) The correlation coefficient
- ☒ (d) The slope of the regression line

$$y - \hat{y}$$

13. The Prosecutor's Fallacy occurs when someone confuses:

- ☒ (a) $P(\text{evidence} \mid \text{innocent})$ with $P(\text{innocent} \mid \text{evidence})$
- ☒ (b) $P(A \text{ or } B)$ with $P(A \text{ and } B)$
- ☒ (c) Mean with median
- ☒ (d) Standard deviation with standard error

* 14. A biased coin has $P(\text{head}) = 0.25$. It is tossed 12 times independently. What is the probability of exactly 4 heads?

- ☒ (a) `choose(12, 4)*(0.75)^4*(0.25)^8`
- ☒ (b) `choose(12, 4)*(0.25)^12`
- ☒ (c) `choose(4, 12)*(0.25)^4*(0.75)^8`
- ☒ (d) `choose(12, 4)*(0.25)^4*(0.75)^8`

$$\begin{aligned} 0.25^4 \cdot 0.75^8 \\ 12 \\ 4 \end{aligned}$$

15. A box model has population mean 6 and population standard deviation 10. If 25 draws are made with replacement, what is the standard error of the sample mean?

- (a) 10
- ☒ (b) 2
- (c) 50
- (d) 6

$$\mu = 6 \quad \sigma = 10$$

$$SE_{\bar{x}} = \frac{SD}{\sqrt{n}} = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2$$

☒ 16. A 95% confidence interval for the proportion of students who prefer online tutorials is (0.18, 0.31). Which interpretation is best?

- ☒ (a) The interval gives plausible values for the population proportion, using a method designed to capture the true parameter in about 95% of repeated samples
- ☒ (b) There is a 95% probability that this fixed interval was generated from the sample
- ☒ (c) Exactly 95% of students in the sample prefer online tutorials
- ☒ (d) The probability that the null hypothesis is true is between 0.18 and 0.31

$$0.18 \xrightarrow{95\%} 0.31$$

17. In a hypothesis test, a p-value of 0.013 means:

0.05

- ☒ (a) There is a 1.3% chance that the null hypothesis is true
- ☒ (b) There is a 98.7% chance that the alternative hypothesis is true
- ☒ (c) If the null hypothesis were true, data at least as extreme as observed would be relatively unlikely
- ☒ (d) The observed statistic is wrong with probability 0.013

18. Consider the R output:

```
t.test(wait_min, mu = 15)
```

One Sample t-test

```
data: wait_min
```

```
t = 2.12, df = 39, p-value = 0.0404
```

```
alternative hypothesis: true mean is not equal to 15
```

```
95 percent confidence interval:
```

```
15.1 18.7
```

```
sample estimates:
```

```
mean of x
```

```
16.9
```

$$p = 0.04$$

$$H_0: \mu = 15, H_A: \mu \neq 15 \quad \alpha = 0.05$$

For $H_0: \mu = 15$ versus $H_A: \mu \neq 15$ at $\alpha = 0.05$, what should be concluded?

- ☒ (a) Do not reject H_0 ; the p-value is larger than 0.05
- ☒ (b) Reject H_0 ; there is evidence that the mean waiting time differs from 15 minutes
- ☒ (c) Accept H_0 ; the sample mean is close to 15
- ☒ (d) Reject H_0 ; there is proof that every waiting time is above 15 minutes

19. A six-sided spinner is claimed to land equally often on each face. A chi-squared goodness-of-fit test is performed. What is the alternative hypothesis?

- ☒ (a) The spinner is fair
- ☒ (b) All six faces have exactly the observed sample proportions
- ☒ (c) At least one face has a population proportion different from $1/6$
- ☒ (d) The p-value equals $1/6$

reading ←

$$\frac{1}{6} \text{ per face}$$

$$H_0: \frac{1}{6} \text{ for every spin}$$

$$H_A: \text{at least 1 spin that isn't } \frac{1}{6}$$

20. Consider the R output:

```
summary(lm(repair_cost ~ age_years, data = cars))
```

```
Coefficients:
```

```
Estimate Std. Error t value Pr(>|t|)
```

$$\widehat{\text{repair cost}} = a + b(\text{age years})$$

$$a = 480.2$$

$$b = 214.6$$

(Intercept)	480.2	110.1	4.36	0.0001
age_years	214.6	58.7	3.66	0.0008

Multiple R-squared: 0.28

Which conclusion is most appropriate from this output alone?

- ☐ (a) Car age causes repair cost because the p-value is small
- ☐ (b) The slope is negative
- ☒ (c) There is evidence of a positive linear association, assuming the model conditions are reasonable
- ☐ (d) $R^2 = 0.28$ means 28% of cars required repairs

Data Story 1: Campus Study Spaces and Student Outcomes

Use the following information to answer Questions 21-30.

A university learning analytics team collects data from 1200 students to explore how study-space use, sleep, stress, and exam performance are related.

```
dim(campus_study)
```

```
[1] 1200 14
```

```
names(campus_study)
```

```
[1] "student_id"      "campus"
[3] "faculty"         "year_level"
[5] "preferred_study_space" "weekly_library_hours"
[7] "online_resource_hours" "sleep_hours"
[9] "paid_work_hours"    "coffee_cups"
[11] "stress_score"      "exam_mark"
[13] "pass"             "quiet_zone_user"
```

$n = 1200$

Variable	Description
student_id	unique student identifier
campus	university campus
faculty	student faculty
year_level	year of study
preferred_study_space	main place used for study
weekly_library_hours	hours per week spent studying in the library
online_resource_hours	hours per week spent using online resources
sleep_hours	average sleep per night
paid_work_hours	paid work hours per week
coffee_cups	average coffees per day
stress_score	stress score from 1 to 10
exam_mark	final exam mark
pass	whether the student passed
quiet_zone_user	whether the student regularly used quiet study zones

Additional R output:

```
first_year = campus_study %>% filter(year_level == "First")
```

```
dim(first_year)[1] / dim(campus_study)[1]
```

```
[1] 0.31
```

31% of students are

```
lm(exam_mark ~ weekly_library_hours, data = campus_study)
```

Coefficients:

```
(Intercept) weekly_library_hours
```

```
58.20
```

```
0.92
```

```
cor(campus_study$weekly_library_hours, campus_study$exam_mark)
```

```
[1] 0.38
```

```
t.test(stress_score ~ quiet_zone_user, data = campus_study)
```

Two Sample t-test

p-value = 0.018

Stress score quiet zone user

sample estimates:

mean in group FALSE mean in group TRUE
6.3 5.7

```
chisq.test(table(campus_study$pass, campus_study$preferred_study_space))
```

Pearson's Chi-squared test

p-value = 0.64

21. How many observations are in the campus_study dataset?

- (a) 14
- ☒ (b) 1200
- (c) 0.31
- (d) 58.20

✓

22. What does it mean that this dataset is tidy?

- (a) It has no missing values
- ☒ (b) Each variable is a column and each student is a row
- (c) It is sorted by exam mark
- (d) All variables are numerical

✓

23. Which variable is most clearly an identifier?

- ☒ (a) student_id
- (b) exam_mark
- (c) stress_score
- (d) pass

✓

24. What does the output 0.31 tell us?

- ☒ (a) 31% of students are first-year students
- (b) 31 first-year students are in the dataset
- (c) 31% of first-year students passed
- (d) First-year students studied for 31 hours per week

✓

25. Which graph would best compare exam marks across faculties?

- (a) Histogram of exam_mark
- (b) Scatterplot of exam_mark against weekly_library_hours
- ☒ (c) Comparative boxplot of exam_mark by faculty
- (d) Bar chart of student_id

✓

26. Which R code would produce a histogram of weekly library hours for first-year students?

- ☒ (a) ggplot(first_year, aes(x = weekly_library_hours)) + geom_histogram()
- ✓ (b) ggplot(first_year, aes(y = weekly_library_hours)) + geom_boxplot()
- ✓ (c) ggplot(campus_study, aes(x = faculty, y = weekly_library_hours)) + geom_point()
- ✓ (d) hist(first_year\$faculty)

27. Which pair of variables is most suitable for simple linear regression?

- ✓ (a) faculty and campus

- ✓ (b) student_id and pass
- Ⓒ weekly_library_hours and exam_mark
- ✓ (d) preferred_study_space and faculty

$$\text{exam mark} = a + b(\text{weekly hrs})$$

58.20 0.92

28. Based on the regression output, which statement is most appropriate?

- Ⓐ Each extra weekly library hour is associated with a predicted 0.92 mark increase
- ~ (b) Each extra weekly library hour ~~causes~~ a 0.92 mark increase
- ✓ (c) The predicted exam mark is always 58.20
- ✓ (d) The correlation is 0.92

29. Using $\alpha = 0.05$, what conclusion follows from the two-sample t-test?

$$\alpha = 0.05$$

- ✓ (a) There is no evidence that quiet-zone use is related to stress score
- Ⓑ There is evidence that quiet-zone users and non-users have different mean stress scores, with quiet-zone users lower in the sample
- ✓ (c) 1.8% of students use quiet zones
- ✓ (d) Quiet-zone use definitely causes lower stress

30. Using $\alpha = 0.05$, what conclusion follows from the chi-squared test?

- ✓ (a) There is evidence that pass status and preferred study space are associated
- Ⓑ There is not enough evidence that pass status and preferred study space are associated
- ✓ (c) Preferred study space causes students to pass
- ✓ (d) The p-value means 64% of students passed

no effect between pass & preferred study sp

29/3

Section B: Extended Answer

There is one question in this section. Demonstrate evidence-based decision making. Your writing should be precise, concise, and supported by the data story and sources.

Data Story 2: Urban Cycling and Commute Safety

A transport analyst is investigating whether cycling infrastructure and route features are associated with safety and perceived safety across Sydney cycling routes. The dataset `cycle_commute` contains 320 route segments.

Source 1: Data Dictionary

Variable	Description
<code>route_id</code>	unique route segment identifier
<code>region</code>	Sydney region
<code>infrastructure_type</code>	main route type: protected lane, shared path, or mixed traffic
<code>protected_route</code>	whether the segment has a protected cycling lane
<code>speed_limit</code>	posted speed limit in km/h
<code>bike_lane_km</code>	kilometres of cycling lane on the route segment
<code>green_cover_pct</code>	percentage of surrounding area with green cover
<code>avg_commute_min</code>	average cycling commute time in minutes
<code>daily_cyclists</code>	estimated number of cyclists per day
<code>incident_rate</code>	reported incidents per 100,000 cycling trips
<code>perceived_safety_score</code>	average safety rating from 1 to 10
<code>high_incident</code>	whether the route segment is classified as high incident or low incident

Source 2: Two-sample t-test

```
t.test(incident_rate ~ protected_route, data = cycle_commute)
```

Two Sample t-test

p-value = 0.0024

95 percent confidence interval:

1.8 5.6

sample estimates:

mean in group FALSE mean in group TRUE
9.7 6.0

Source 3: Chi-squared test

```
chisq.test(table(cycle_commute$region, cycle_commute$high_incident))
```

Pearson's Chi-squared test

X-squared = 4.36

```
df = 4
p-value = 0.36
```

Source 4: Linear model

```
summary(lm(perceived_safety_score ~ bike_lane_km, data = cycle_commute))
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.10	0.41	10.00	<0.001
bike_lane_km	0.085	0.029	2.93	0.004

Residual standard error: 1.8

Multiple R-squared: 0.12

Question 1

(a) A draft council report says:

“Building bike lanes causes all Sydney commuters to feel safer, especially in Western Sydney, and will eliminate cycling incidents.”

Explain two problems with this statement using the sources above.

(b) A student writes the following incorrect explanation of Source 3:

“The p-value is 0.36, which means 36% of regions are high incident.”

Write a correct explanation of the Source 3 result.

(c) The analyst wants to investigate the claim:

“Routes with higher speed limits tend to have higher incident rates.”

Describe an appropriate plot and write R code that could produce it.

(d) Suggest one new research question using three variables from Source 1 that are not all already used together in Sources 2-4.

(e) For your research question in part (d), briefly explain what statistical method or graphical summary would be appropriate.

Practice Exam Guide

This guide summarises formulas and R commands that are useful for this practice exam. It is based on the local DATA1901 exam guide and mastery notes.

Numerical Summaries

Idea	Formula or rule
IQR	$IQR = Q3 - Q1$
Boxplot lower threshold	$LT = Q1 - 1.5 \cdot IQR$
Boxplot upper threshold	$UT = Q3 + 1.5 \cdot IQR$
Linear transformation	If $y = a + b \cdot x$, then $\text{mean}(y) = a + b \cdot \text{mean}(x)$ and $SD(y) = \text{abs}(b) \cdot SD(x)$

Models

Model	Key idea
Normal	$X \sim N(\text{mean}, SD^2)$; use <code>pnorm(value, mean, SD)</code> for lower-tail areas
Binomial	$P(X = x) = \text{choose}(n, x) p^x (1-p)^{(n-x)}$
Box model sample mean	$EV(\text{mean}) = M, SE(\text{mean}) = SD/\sqrt{n}$
Regression	Slope is predicted change in response for a one-unit increase in explanatory variable

Hypothesis Testing

HATPC means:

1. Hypotheses
2. Assumptions
3. Test statistic
4. P-value
5. Conclusion in context

Common tests:

Situation	Test
One binary outcome vs a constant	One-sample proportion test
One mean vs a constant, population SD unknown	One-sample t-test
Two independent means	Two-sample t-test
One categorical variable vs a model	Chi-squared goodness-of-fit test
Two categorical variables	Chi-squared independence test
Two quantitative variables	Regression slope test

R Commands

```
library(tidyverse)
ggplot(data, aes(x = quantitative_var)) + geom_histogram()
ggplot(data, aes(x = group_var, y = quantitative_var)) + geom_boxplot()
```

```
ggplot(data, aes(x = x_var, y = y_var)) + geom_point()
pnorm(value, mean, sd)
qnorm(probability, mean, sd)
pbinom(x, size, prob)
lm(y ~ x, data = data)
t.test(y, mu = constant)
t.test(y ~ group, data = data)
chisq.test(table(var1, var2))
```

Answer Key

Multiple Choice Answers

Question	Answer	Question	Answer	Question	Answer
1	A	11	A	21	B
2	B	12	B	22	B
3	B	13	A	23	A
4	C	14	D	24	A
5	B	15	B	25	C
6	D	16	A	26	A
7	C	17	C	27	C
8	B	18	B	28	A
9	D	19	C	29	B
10	C	20	C	30	B

Worked Solutions

Multiple Choice Worked Solutions

Q1. Answer: A

Data provenance is the record of the origin and processing history of data. It is about trust, context, and interpretation, not probability or graph design.

Q2. Answer: B

A randomised controlled trial requires the researcher to assign units to treatment conditions at random. Option B has random assignment to app or usual habits.

Q3. Answer: B

A random sample can support generalisation to the sampled population. Because no treatment was assigned, the study cannot by itself prove a causal effect.

Q4. Answer: C

One qualitative variable and one quantitative variable are commonly displayed using comparative boxplots, with the qualitative variable defining groups.

Q5. Answer: B

Tidy data has variables in columns and observations in rows. Tidy does not guarantee no missing values or that variables are numerical.

Q6. Answer: D

For $y = 4 - 0.5x$, the mean is $4 - 0.5 \cdot 12 = -2$. The SD is $\text{abs}(-0.5) \cdot 5 = 2.5$; SD cannot be negative.

Q7. Answer: C

$IQR = 30 - 18 = 12$. The upper threshold is $30 + 1.5 \cdot 12 = 48$. A high outlier must be above 48, so 49 is the first listed high outlier.

Q8. Answer: B

$X \sim N(80, 4^2)$ has mean 80 and SD 4. A between probability is upper lower-tail area minus lower lower-tail area: $\text{pnorm}(84, 80, 4) - \text{pnorm}(72, 80, 4)$.

Q9. Answer: D

Chance error is random variation in repeated measurements under the same conditions. The SD of repeated measurements estimates its typical size.

Q10. Answer: C

Correlation is unchanged by shifts and positive scaling. Multiplying one variable by a negative number reverses the sign. Only x is negatively scaled, so -0.60 becomes 0.60 .

Q11. Answer: A

In $\text{energy_use} \sim \text{temperature}$, the response is energy use and the explanatory variable is temperature. The slope is -0.42 , so predicted energy use decreases by 0.42 for a one-unit temperature increase.

Q12. Answer: B

A residual is observed y - predicted y . It is the vertical gap from the point to the fitted line.

Q13. Answer: A

The Prosecutor's Fallacy confuses the probability of observing evidence if innocent with the probability of innocence given the evidence. These are conditional probabilities in opposite directions.

Q14. Answer: D

Binomial probability for exactly 4 heads in 12 independent tosses is $\text{choose}(12, 4) * (0.25)^4 * (0.75)^8$. The exponent on failure is $12 - 4 = 8$.

Q15. Answer: B

For a sample mean from a box model, $SE(\text{mean}) = SD/\sqrt{n} = 10/\sqrt{25} = 10/5 = 2$.

Q16. Answer: A

A confidence interval is a range of plausible parameter values produced by a long-run reliable method. It is not the probability that H_0 is true and not a statement that exactly 95% of sampled students prefer online tutorials.

Q17. Answer: C

A p-value is calculated under the null hypothesis. A value of 0.013 says that data at least this extreme would be uncommon if H_0 were true.

Q18. Answer: B

The p-value 0.0404 is less than 0.05, and the 95% CI excludes 15. Reject H_0 ; there is evidence that the mean waiting time differs from 15 minutes.

Q19. Answer: C

For a chi-squared goodness-of-fit test, the null says the model proportions hold. The alternative says at least one true category proportion differs from the claimed model.

Q20. Answer: C

The slope estimate is positive and its p-value is small, so there is evidence of positive linear association if model assumptions are reasonable. This does not prove causation, and $R^2 = 0.28$ means the model explains about 28% of variation in repair cost, not that 28% of cars were repaired.

Q21. Answer: B

`dim(campus_study)` gives rows then columns. The output `[1] 1200 14` means 1200 observations and 14 variables.

Q22. Answer: B

In this data story, one observation is one student. Tidy data means each student is a row and each variable is a column.

Q23. Answer: A

`student_id` is the unique identifier. It labels the record but is usually not a meaningful quantitative variable for analysis.

Q24. Answer: A

The code divides the number of first-year rows by the total number of rows. 0.31 means about 31% of students are first-year students.

Q25. Answer: C

To compare a quantitative variable (`exam_mark`) across categories (`faculty`), use comparative boxplots.

Q26. Answer: A

A histogram of a quantitative variable maps that variable to `x` and uses `geom_histogram()`. The data should be `first_year` because the question asks for first-year students.

Q27. Answer: C

Simple linear regression uses two quantitative variables. `weekly_library_hours` and `exam_mark` are both quantitative and form a plausible explanatory-response pair.

Q28. Answer: A

The slope 0.92 means that each extra weekly library hour is associated with a predicted 0.92 mark increase. Because the data are observational, causal language is not justified.

Q29. Answer: B

The p-value 0.018 is less than 0.05, so there is evidence of different mean stress scores. The sample means show quiet-zone users have lower mean stress score (5.7 vs 6.3).

Q30. Answer: B

The chi-squared p-value 0.64 is much larger than 0.05, so there is not enough evidence of association between pass status and preferred study space.

Extended Answer Model Solution

1(a) Problems with the draft council report

The statement was:

“Building bike lanes causes all Sydney commuters to feel safer, especially in Western Sydney, and will eliminate cycling incidents.”

There are several problems. A high-quality answer only needed two, but the strongest answers would identify more than two.

Problem 1: The statement uses causal language.

Source 4 shows an association between `bike_lane_km` and `perceived_safety_score`. The slope is positive (0.085) and the p-value is small (0.004), so there is evidence that routes with more bike-lane kilometres tend to have higher perceived safety scores. However, this is observational route data. The output does not show random assignment of bike lanes to routes, so it does not prove that building bike lanes causes higher perceived safety.

Problem 2: The statement says “all Sydney commuters”.

Source 4 gives an average linear relationship. It does not say every commuter or every route segment has higher perceived safety. The R^2 value is only 0.12, so bike-lane kilometres explain only part of the variation in perceived safety scores.

Problem 3: The statement says “especially in Western Sydney”.

None of Sources 2-4 tests whether the relationship differs by region or specifically by Western Sydney. Source 3 uses `region`, but it tests `region` against `high_incident`, not whether bike-lane effects differ by region.

Problem 4: The statement says bike lanes “will eliminate cycling incidents”.

Source 2 shows lower sample mean incident rate for protected routes (6.0) than non-protected routes (9.7) and a p-value of 0.0024, which is evidence of a difference in mean incident rates. But the incident rate for protected routes is not zero, and the test does not support the word “eliminate”.

An exemplary concise answer:

The statement overclaims. Sources 2 and 4 show associations, not causal effects, because there is no randomised experiment assigning infrastructure to routes. It also overgeneralises: Source 4 is an average relationship with $R^2 = 0.12$, not proof that all commuters feel safer. The “especially in Western Sydney” claim is unsupported because no source tests an interaction with region or a Western Sydney subgroup.

1(b) Correct explanation of Source 3

Source 3 is a chi-squared test of independence between `region` and `high_incident`.

The null hypothesis is that `region` and `high_incident` are independent. The alternative hypothesis is that they are associated.

The p-value is 0.36, which is larger than common significance levels such as 0.05. Therefore, the data do not provide enough evidence to reject independence. In context:

There is not enough evidence, based on this dataset, that high-incident classification differs by Sydney region.

The p-value does not mean that 36% of regions are high incident. It is a tail probability calculated under the null hypothesis.

1(c) Appropriate plot and R code

The claim compares two quantitative variables:

- speed_limit: quantitative explanatory variable
- incident_rate: quantitative response variable

An appropriate plot is a scatterplot of incident_rate against speed_limit.

Example R code:

```
library(tidyverse)

ggplot(cycle_commute, aes(x = speed_limit, y = incident_rate)) +
  geom_point() +
  labs(
    title = "Incident rate by speed limit",
    x = "Speed limit (km/h)",
    y = "Incidents per 100,000 cycling trips"
  )
```

A stronger version could colour by infrastructure_type to check whether the pattern differs across route types:

```
ggplot(cycle_commute,
       aes(x = speed_limit,
           y = incident_rate,
           colour = infrastructure_type)) +
  geom_point() +
  labs(
    title = "Incident rate by speed limit and infrastructure type",
    x = "Speed limit (km/h)",
    y = "Incidents per 100,000 cycling trips"
  )
```

1(d) New research question using three variables

One suitable research question is:

Is the relationship between daily cyclist volume and incident rate different across infrastructure types?

This uses three variables:

- daily_cyclists
- incident_rate
- infrastructure_type

These three variables are not all already used together in Sources 2-4.

Other valid examples include:

- Is the relationship between green cover and perceived safety different across regions?
- Are commute times related to incident rates, and does this differ between protected and non-protected routes?
- Do routes with higher speed limits have different incident rates depending on infrastructure type?

1(e) Appropriate method or graphical summary

For the proposed question:

Is the relationship between daily cyclist volume and incident rate different across infrastructure types?

Both `daily_cyclists` and `incident_rate` are quantitative variables, while `infrastructure_type` is categorical. A suitable first analysis is a scatterplot with colour showing infrastructure type:

```
ggplot(cycle_commute,
       aes(x = daily_cyclists,
           y = incident_rate,
           colour = infrastructure_type)) +
geom_point() +
labs(
  title = "Incident rate by cyclist volume and infrastructure type",
  x = "Daily cyclists",
  y = "Incidents per 100,000 cycling trips"
)
```

If the scatterplot looks roughly linear within groups, a regression model could be considered. The interpretation should remain associational unless the study design supports causation.

Marking Guide for Extended Answer

Part	Marks	Examiner expectations
1(a)	8	Identifies two genuine problems; explains causation vs association; avoids unsupported subgroup claims; uses Sources 2-4 accurately.
1(b)	5	Names the chi-squared independence test; states correct null/alternative meaning; interprets p-value correctly; concludes in context.
1(c)	5	Identifies both variables as quantitative; chooses scatterplot; gives syntactically plausible ggplot code; labels axes in context.
1(d)	3	Gives a coherent research question using three variables from Source 1; avoids merely repeating Sources 2-4.
1(e)	4	Matches method/graph to variable types; explains why it fits; optionally gives R code; avoids causal overclaim.

Examiner Pattern Notes

This practice exam deliberately targets common DATA1901 assessment patterns:

Pattern	Where it appears
Study design limits conclusions	Q2, Q3, Q28, Extended 1(a)
Variable type determines graph/test	Q4, Q25-Q27, Extended 1(c), 1(e)
R output interpretation	Q11, Q18, Q20, Q21-Q30
Normal/binomial/box model syntax	Q8, Q14, Q15
p-value interpretation	Q17-Q19, Q29-Q30, Extended 1(b)
Association vs causation	Q3, Q20, Q28, Q29, Extended 1(a), 1(e)
Confidence interval interpretation	Q16, Q18, Extended Source 2
Data-story reasoning	Q21-Q30, Extended Answer

High-scoring answers usually do three things:

1. Name the statistical object correctly: variable type, parameter, statistic, model, or test.
2. Interpret the result in context.
3. Avoid overclaiming beyond the study design.